PRD: Mirage

The Anti-Adversarial Copy-Trading Intelligence Layer for Four.meme

Date: April 19, 2026 **Domain:** AI x Crypto **Confidence Score:** 56 / 60 **Generated via:** /generate-prd **Author:** Mirage Product Team **Status:** Build Phase — Four.meme AI Sprint (submission April 22, 2026)

Cover Paragraph

Mirage is an anti-adversarial intelligence layer for memecoin copy-trading on Four.meme and BNB Chain. It uses a multi-agent LLM system with chain-of-thought reasoning to analyze wallets and tokens, producing explainable **Trust Verdicts** (Copy / Avoid / Uncertain) backed by human-readable evidence. Built on state-of-the-art 2026 academic research (Luo et al., arXiv 2601.08641), Mirage is the first tool to directly attack the problem that most "smart money" wallets shilled to retail are actually adversarial bots farming copy-traders. Surfaces include a free Telegram bot (growth engine), a freemium web dashboard (primary trader interface), optional auto-copy execution (monetization), and creator self-audit (secondary audience). The hackathon deliverable is a working product, not a prototype — demo-able live with verifiable catches on known-adversarial wallets.

Pitch in one sentence: *Every other tool tells you who's winning. Mirage tells you who's cheating.*

Idea Scoring Matrix

The narrative anchors this PRD, but per /generate-prd protocol, here is the scoring of Mirage against the next-best alternative concepts a hackathon team could pursue in the same window.

Idea	Timing	Demand	Comp Gap	Feasibility	Monetization	Scope	Tota
Mirage — Anti-adversarial copy-trading intel (BNB/Four.meme)	10	10	10	8	9	9	**56**
Generic BNB rug-pull detector	7	7	4	9	6	6	39
AI memecoin creator/persona bot	6	7	3	8	6	7	37
BNB copy-trade auto-executor (no intel layer)	7	8	5	7	8	7	42
On-chain wallet reputation NFT protocol	5	5	7	4	5	6	32

Selection: Mirage. Highest across every criterion, is the single idea that productizes a peer-reviewed 2026 paper, and lives cleanly in the AI x Crypto intersection.

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1. Executive Summary

Hundreds of thousands of retail traders open Telegram bots (GMGN, Axiom, BullX, MEVX, WalletFinder, KOLScan) every day and copy "smart money wallets." They lose — because most of those wallets are not smart money. They are adversarial bots built to farm copy-traders via bundle coordination, sniper positioning, wash trading, and social manipulation. The January 2026 ACM WWW paper by Luo et al. proved this across 6,000+ memecoin projects and proved that multi-agent LLMs with chain-of-thought reasoning can detect it.

This problem is most acute on **BNB Chain**. Four.meme has reached 812,000 daily unique users and $1B+ trading volume in PancakeSwap-listed pairs, yet every copy-trading tool in market is Solana-first. BNB traders are flying blind, copy-trading with Solana-shaped tools against BNB-shaped adversaries.

Mirage is the fix. It does not display wallets — it *reasons* about them. For every Four.meme wallet and token, three specialized LLM agents (Coin, Wallet, Timing) evaluate structured on-chain features, produce a Trust Verdict (Copy / Avoid / Uncertain), and return a natural-language reasoning trace the user can read, audit, and disagree with.

Why now: (1) Four.meme just crossed $1B in volume and announced a $50K AI Sprint (April 1–30, 2026); (2) the academic research proving the approach works was published at WWW 2026; (3) no shipping product has productized the research; (4) the BNB chain gap is wide open. Four weeks is enough to ship a working product *and* demo it live at the hackathon.

6-month target: 10,000+ Telegram bot users, validated accuracy against a labeled adversarial dataset, first paying premium subscribers, active integration conversations with Four.meme.

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2. Market Context & Research Backing

Trends driving this opportunity

- **BNB memecoin explosion.** Four.meme: 812K DAU, $1B+ PCS volume, 52K tokens, 27K creators, >33K BNB cumulative protocol revenue (BNB Chain Blog).
- **Copy-trading adoption.** GMGN alone hit $100M+ daily volume and ~50% of Solana memecoin flow by April 2025 (GMGN 2026 guide). Copy-trading is a core workflow, not a niche.
- **Research maturity.** Luo et al., *"Resisting Manipulative Bots in Meme Coin Copy Trading: A Multi-Agent Approach with Chain-of-Thought Reasoning,"* ACM WWW 2026, arXiv:2601.08641. Multi-agent CoT beats zero-shot LLMs and statistical baselines; achieves 3% average copier return per memecoin under realistic frictions.
- **Hackathon timing.** Four.meme AI Sprint with $50K prize pool, build window April 8–21, submission April 22 (DoraHacks).

Market size

- **TAM (global memecoin copy-trading tools):** $5–10B annual revenue opportunity on ~$1T+ annualized memecoin DEX flow.
- **SAM (BNB-specific trader tooling):** ~$30–90M annual, based on 100–300K active BNB meme traders at $100–300 annual ARPU.
- **SOM (18 months):** $2–8M ARR realistic, from 5–10% of SAM across Telegram, dashboard, and API surfaces.

Growth rate of BNB meme segment: 3–5× year-over-year as of Q1 2026 (derived from cumulative-token and DAU trajectories reported on DappBay and BNB Chain blog).

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3. Problem Statement

Target persona — Ravi, the BNB copy-trader

- **Profile:** 22–38, crypto-native, trades from Telegram and Twitter more than from exchanges.
- **Workflow:** Wakes, opens Telegram alpha groups, sees "this wallet is 287% PnL last 30 days, up 94% win rate, here is the address," loads into GMGN or BullX, enables one-click copy. Rinse 10–30 times a day.
- **Budget:** $500–$5,000 rotating across memecoin bets. Losses compound quickly.

Current pain (specific, measurable)

- **Hit rate on copy-traded "alpha" wallets: consistently negative.** Bundle bots, snipers, wash-traders pose as smart money. Retail enters after the insiders are already distributing.
- **No way to verify.** Existing tools rank wallets by PnL and win rate — the *exact* metrics adversarial bots are engineered to optimize.
- **Time cost of manual forensics:** 20–40 minutes per wallet to inspect block-level buy clustering, funding ancestors, cross-token variance. Nobody does it.

Current "solutions" and their failure modes

Tool	What it does	Why it fails Ravi
GMGN	Telegram copy-trade + wallet ranking	Multi-chain but Solana-first; ranks by PnL; no adversary detection.
Axiom	Web terminal, scanners, wallet/X tracking	Solana-only; premium; no adversary reasoning.
BullX	Hybrid Telegram/web, pay-per-trade	Solana-first; same PnL-based ranking.
MEVX / WalletFinder / KOLScan	Wallet display tools	Display-only; no reasoning; adversarial wallets appear as alpha.
RugCheck / TokenSniffer / VibeCheck	Token-level safety	Token-only; silent on wallet trust.

The gap

No tool on BNB Chain reasons about whether a "smart money" wallet is itself adversarial. The category has no incumbent. The science for solving it is published. The distribution channel (Telegram) is primed. The gap is shipping-product-shaped and Four-weeks-wide.

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4. Proposed Solution

Product description

Mirage is a multi-agent reasoning engine wrapped in two primary surfaces — a free Telegram bot and a freemium web dashboard — that takes any Four.meme wallet or token and returns an explainable Trust Verdict: **Copy**, **Avoid**, or **Uncertain**, with a composite 0–100 Trust Score, three reasoning traces, cited on-chain evidence, and a counter-argument.

Core value proposition

"We don't rank wallets. We reason about them — and show you our work."

Key differentiators

1. **BNB-native by design.** The only serious copy-trade intelligence product built for BNB Chain, not ported from Solana.
2. **Reasoning is the product.** Every verdict ships with a human-readable chain of thought — a trader can audit and override.
3. **Academically grounded.** Direct productization of ACM WWW 2026 research, not vibes or hand-tuned heuristics.
4. **Evidence-cited.** Every reasoning claim is bound to a specific block, tx hash, or wallet address — hallucinations are rejected by a post-processing validator.
5. **Counter-argument included.** Every verdict explains what would need to be true to invalidate it — a trust-building feature no competitor offers.

Why this is 10× better, not 10% better

Existing tools answer "*which wallets have the highest PnL?*" Mirage answers the fundamentally different question "*is this wallet farming me?*" The first question has been rewarded by adversaries who optimize for it; the second question has no incumbent. Moving from "display" to "reasoning" is a category jump, not an improvement increment.

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5. Key Features (MVP Scope)

P0 — Must-have (hackathon submission)

Feature	Description	Effort
F1 — Wallet Trust Verdict API	`POST /analyze_wallet` returns JSON verdict + reasoning trace + evidence citations for any BNB address.	5 eng-days
F2 — Token Trust Verdict + Ranked Buyers	`POST /analyze_token` returns token verdict, bundle-bot contamination score, and buyer table sorted by Trust Score.	5 eng-days
F3 — Telegram Bot	`/analyze <wallet|token>` with inline expand for reasoning trace. Shareable screenshot-ready output.	4 eng-days
F4 — Web Dashboard	Next.js + shadcn/ui: paste token address → token card + buyer table + signal strength + expand-for-reasoning.	6 eng-days
F5 — Three-Agent Reasoner	LangGraph orchestration of Coin, Wallet, Timing agents with few-shot CoT prompts and evidence-cite enforcement.	6 eng-days
F6 — Feature Pipeline	BscScan + Dune ingestion, Redis-cached features for every active Four.meme wallet in last 30 days.	7 eng-days

P1 — Should-have (if Week 2 allows)

Feature	Description	Effort
F7 — Wallet Subscription Pings	User subscribes to a trust-verified wallet; Telegram alert on new buy with organic/engineered read.	3 eng-days

F8 — Creator Self-Audit	Creator submits token; gets free report on bundle contamination in first 50 buyers.	2 eng-days
F9 — Historical Backtest Page	Static page showing Mirage's verdicts on 50 known-historical scams — builds judge trust.	2 eng-days

P2 — Post-hackathon

- Non-custodial session-key auto-copy execution
- Multi-chain expansion (Solana, Base)
- Public API (priced) for integrations
- Decentralized attestation of Trust Verdicts (optional on-chain anchor)

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6. Technical Architecture

Stack

Layer	Choice	Justification
Data ingestion	BscScan API, Dune Analytics (sql), LunarCrush lightweight scraper	Covers tx-level, historical, and social signals respectively. All accessible within hackathon free tiers.
Feature store	Postgres (managed, Supabase) + Redis (Upstash)	Warm cache for sub-second queries; durable historical features in Postgres.
Orchestration	LangGraph (Python)	Native multi-agent graphs; aligns with Luo et al. reference design.
LLM inference	Llama 3.3 70B or Qwen 2.5 72B via Modal/Together	Open-weights sufficient for structured-feature CoT; predictable cost (~$0.001/trace).
Telegram bot	grammY (TypeScript) on Fly.io	Fastest Telegram framework; simple deploy.
Web	Next.js 15 + shadcn/ui + Vercel	Fastest UX ship; designer-quality default components.
Monitoring	Vercel Analytics, PostHog	Funnel and retention tracking.

Architecture (ASCII)

```
┌──────────────────────────┐
│   User Surfaces          │
│  Telegram Bot | Web UI   │
└──────────────────────────┘
              │
              │ REST / WebSocket
┌──────────────▼───────────┐
│   Mirage API (Next.js /  │
│   Fastify edge)          │
└──────────────────────────┘
              │
              │
┌──────────────▼───────────┐
```

```
                    │  Trust Verdict Engine     │
                    │   composes agent outputs  │
                    └───────┬────────┬─────────┘
                            │        │         │
                     ┌──────▼──┐ ┌──▼──────┐ ┌─▼────────┐
                     │ Coin    │ │Wallet   │ │Timing    │
                     │ Agent   │ │ Agent   │ │ Agent    │
                     └─────────┘ └─────────┘ └──────────┘
                            └────────┬────────┘
                                     │ structured features
                    ┌────────────────▼──────────────────┐
                    │   Feature Store                   │
                    │   Redis (hot) + Postgres          │
                    └───────────────────────────────────┘
                                     │
                    ┌────────────────▼──────────────────┐
                    │   Ingestion Workers               │
                    │   BscScan | Dune | Social         │
                    └───────────────────────────────────┘
```

Data flow

1. Worker pool continuously ingests every Four.meme tx → Postgres raw store.
2. Feature extractor recomputes structural features (bundle coefficients, co-buyer Jaccard, block-timing entropy, graph distances, cross-token alpha variance) on a rolling 30-day window → Redis.
3. User requests wallet/token → API reads cached features → sends to three agents in sequence → Trust Verdict Engine composes output.
4. Response streams to surface with reasoning trace expandable.

Infrastructure cost estimate

Item	Monthly (hackathon scale)
Modal / Together inference	$100–300
Postgres (Supabase Pro)	$25
Redis (Upstash)	$20
Vercel + Fly.io	$50
BscScan paid tier (if needed)	$99
Total	**~$300–500 / month**

Security considerations

- No custody of user funds at MVP.
- Telegram bot uses read-only user IDs; no private keys.
- Auto-copy (P2) uses non-custodial session keys with user-approved spending caps.
- All LLM traces pass an evidence-citation validator before display — prevents hallucinated block numbers.
- Rate limits on free tier to prevent scraping.

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7. User Flow

Primary journey — Retail copy-trader on Telegram (6 steps)

1. **Discovery:** User sees a "Mirage AVOID" screenshot in an alpha Telegram group (organic growth loop). Opens @MirageBot.
2. **Onboarding (30 seconds):** Bot greets with a single-message explainer + "Paste any BNB wallet or Four.meme token to test me." No account needed.
3. **First query:** User pastes wallet `0xabcd…`. Bot returns within 2–3s: verdict card + 3 reasoning bullets + "Tap to see full chain of thought."
4. **Expand:** User taps — reads Coin Agent → Wallet Agent → Timing Agent traces. Each claim has a block number and tx hash.
5. **Decide:** User either copies (verdict: Copy) or skips (verdict: Avoid). Optionally taps "Counter-argument" to see what would invalidate the verdict.
6. **Subscribe:** User subscribes to a verified-alpha wallet; receives Telegram pings on future buys with fresh timing verdicts.

Secondary journey — Active trader on Web dashboard

1. Trader pastes a fresh Four.meme token address.
2. Token card loads: overall verdict, creator history, liquidity health, bundle contamination %, graduation probability.
3. Buyer table loads: every wallet that bought in first N minutes, sorted by Trust Score.
4. Trader expands top 3 trust-verified buyers — reads reasoning, checks if multiple trust-verified wallets clustered → signal strength card turns green.
5. Trader executes manually (or via P2 auto-copy).
6. Exit watchdog monitors copied wallets for exit-engineering behavior; pings when distribution phase detected.

Critical screens (ASCII)

```
┌─ Telegram verdict card ──────────────────────────┐
│  ⚠️  AVOID — Trust Score 12/100                   │
│  0xabcd...1234                                    │
│                                                   │
│  • 94% of buys bundled with 4-wallet cluster      │
│  • Funded by wallet that rugged 4 tokens / Mar '26 │
│  • Zero cross-token alpha — dumps within 11 min   │
│                                                   │
│  [ ↘ See full reasoning ]   [ ⊘ Counter-argument ] │
│  [ 🔔 Subscribe ]           [ 🔗 Share ]          │
└──────────────────────────────────────────────────┘
```

```
┌─ Web dashboard token card ───────────────────────┐
│  Token: $MOON  |  🟡 UNCERTAIN — Score 54/100     │
│  Creator: rug-history 1 / legit 3 | Liquidity 🟢  │
│  Bundle contamination: 18% (blocks 42891334–36)   │
│  Graduation probability: 41%                      │
│                                                   │
│  ─── Ranked early buyers (first 8 min) ───        │
│  Wallet            Trust   PnL    Age   Expand    │
│  0x11..  ✅ 87   +412%  220d    [↘]              │
```

```
│  0x22..  ⚠️  09   +890%  3d      [↘]              │
│  0x33..  ✅ 74   +120%  180d    [↘]              │
│  0x44..  ⚠️  14   +612%  8d      [↘]              │
└──────────────────────────────────────────────────┘
```

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8. Monetization Strategy

Revenue model: freemium with premium tier + API tier

Tier	Price	Features
Free (Telegram + Web)	$0	10 token/wallet analyses per day, read-only, shareable screenshots. Growth engine.
Pro	$20 / month	Unlimited analyses, wallet watchlists, subscription alerts, priority queue, historical backtests.
Alpha	$99 / month	Pro + exit watchdog, creator self-audit pro, programmatic CSV export.
API	$299 / month (1M calls)	Programmatic access for other tools and agents.
Auto-Copy (post-MVP)	10% of profits or $49/mo flat	Non-custodial execution with session keys and stop-losses.

Pricing benchmarks

- GMGN: no subscription, trade-fee model.
- BullX: ~1% per trade.
- Axiom: 0.75–0.95% per trade.
- Photon / Banana Pro: trade-fee-based.

Mirage uses a **subscription** model because the value is decision-support, not execution. This is deliberate differentiation — it means Mirage remains useful even when the user trades elsewhere. It also produces a cleaner unit economics story.

Unit economics (target, steady state)

- CAC (Telegram organic + content): **$3–5**
- LTV (Pro tier, 8-month average retention): **$160**
- Payback period: **< 1 month**
- Gross margin on subscription tier: ~80% (inference is the only significant cost)

Revenue projection

Milestone	Users	Paid conv	Monthly rev
Month 3	5,000 TG	2% Pro	~$2,000
Month 6	20,000 TG + 3,000 dashboard	3.5%	~$21,000
Month 12	80,000 TG + 15,000 dashboard	4%	~$100,000–150,000

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9. Go-to-Market Strategy

Pre-launch (April 8–21, 2026 — hackathon build window)

- Invite 50 BNB memecoin traders from Telegram alpha groups to private beta.
- Curate 20 known-adversarial and 20 known-legit wallets as demo set.
- Seed 5 "Mirage caught this" screenshots in 3 target Telegram groups during build week.

Launch day (April 22, 2026 — submission day)

- Submit to Four.meme AI Sprint on DoraHacks.
- Publish launch thread on X anchored to a *live wallet catch* of a popular but adversarial "alpha wallet."
- Post submission video to r/CryptoCurrency, r/BNBChain, and 4 memecoin Discords.
- DM a handful of BNB influencers with the "AVOID" screenshot of a wallet they previously promoted.

Month 1–3 growth (ranked by expected ROI)

1. **Organic screenshot loop** (highest ROI): every AVOID verdict is built to be screenshot-shared. Anti-scam content dominates crypto X/Telegram virality.
2. **Four.meme partnership path**: offer free creator self-audit; position Mirage as an ally to honest creators.
3. **Integration with existing trader tools**: propose API embed in Banana Pro, Photon, or MEVX.
4. **Weekly "rogues gallery"**: publish top 10 adversarial wallets caught this week — content engine.
5. **Seed content on major crypto YouTube channels**: 90-second demo format.

Month 3–6 scaling

- Launch paid Pro tier.
- Ship Solana coverage (same engine, new feature adapter) — extends TAM.
- Partner with 1 DEX aggregator on reverse pre-trade hook: "Mirage score shown before swap."
- Begin enterprise tier conversations with large Telegram alpha services.

Partnerships to pursue

- **Four.meme** (self-audit tool, exclusive launch integration)
- **BNB Chain** (ecosystem grant, data partnership)
- **PancakeSwap** (pre-swap Mirage score badge)
- **Dune Analytics** (data-partner dashboard)
- **LunarCrush** (social-signal feed)

Content / SEO

- SEO anchor pages: "how to spot bundle bots on BNB," "why GMGN smart money fails on BNB," "Four.meme token audit tool."
- Weekly research post on caught adversaries (leveraging the data we already produce).

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10. Competitive Landscape

Direct competitors

Product	Chains	Intel	UX	Pricing	Moat
GMGN	SOL+ (BNB supported,	PnL ranking only	TG + web	Trade fees	Volume + habit

	Solana-first)				
Axiom	SOL + Hyperliquid	Advanced scanners, no adv detection	Web terminal	0.75–0.95%	UX + speed
BullX	SOL-first	PnL ranking	Hybrid TG/web	~1%	User base
MEVX / Photon / Banana Pro	SOL primarily	Display-only	Web	Trade fees	Scanner quality
WalletFinder / KOLScan	SOL	Display-only	Web	Free	SEO
RugCheck / TokenSniffer	Multi	Token-only	Web	Freemium	Scanner depth

Mirage's wedge: the one axis no competitor occupies — *reasoning-grounded wallet trust on BNB Chain with explainable output.*

Positioning map (2×2)

```
                  Display-only  |  Reasoning-based
                                |
SOL-first    GMGN, Axiom        |
             BullX, Photon      |     (empty)
             MEVX,              |
             WalletFinder       |
-------------------------------|-------------------
                                |
BNB-native   MEVX BNB mode,     |     ⋆ MIRAGE ⋆
             GraFun             |
                                |
```

Indirect competitors

- **RugCheck** / **TokenSniffer** — token-level risk, silent on wallets.
- **Nansen** / **Arkham** — wallet labeling, but expensive, institutional, not memecoin-tuned, no CoT.
- **VibeCheck** (prior BNB hackathon winner) — contract heuristics, not wallet trust.

SWOT

Strengths — Only product on BNB with CoT reasoning; academic-grade foundation; free Telegram surface = built-in virality; evidence-cited output = unique trust.

Weaknesses — LLM inference cost at scale (mitigated by open-weights + caching); adversarial adaptation (mitigated by structural features); not a one-click execution tool yet.

Opportunities — Four.meme partnership; expand to Solana/Base with same engine; creator-side paid audit; pre-swap embed in DEX aggregators; API for other tools.

Threats — GMGN/Axiom ship a competing intel layer; Four.meme builds in-house; regulatory classification of verdicts; adversary operations shift chains.

Defensibility / moat

- **Data moat.** Every query adds to a labeled corpus of adversarial/legit wallets. Corpus compounds.
- **Network effect.** Wallet subscriptions produce alert volume that trains routing and prioritization — flywheel.
- **Reputation / brand.** In crypto, the first product that credibly catches scams owns the niche — see RugCheck on Solana.
- **Switching cost.** Once a user has a watchlist, historical verdicts, and notification routing tuned, churn drops.

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11. Risk Assessment

Risk	Severity	Likelihood	Mitigation
LLM hallucinates evidence	High	Medium	Agent prompts forbid unsourced claims; post-processing validator rejects traces referencing off-vector facts. Every claim must cite block number or tx hash.
Feature pipeline too slow for realtime	Medium	Medium	Rolling pre-compute of features for all active Four.meme wallets. User queries hit Redis cache. Sub-second latency achievable.
Adversaries adapt to Mirage signals	Medium	High (long-term)	Structural features (funding graph, cross-token variance) force adversaries to change operations, not just cosmetics — shrinks their edge. Add features quarterly; moving defense.
LLM inference cost at scale	Medium	Low near-term	Open-weights models on Modal/Together; $0.001/trace. Free-tier rate limits; premium subsidizes. Re-evaluate at 100K DAU.
Four.meme builds in-house	Medium	Medium	Mirage's independence is a feature — users trust third-party intel over platform's self-assessment. Pursue partnership not competition.
Regulatory scrutiny of "verdicts"	Medium	Low near-term	Frame as "information, not financial advice." Terms of service explicit. No custody at MVP.
GMGN/Axiom ship intel layer	High	Medium-Low	Speed advantage: Mirage is first. Brand advantage: reasoning-first positioning. Partnership optionality: Mirage is neutral, could be the intel provider to every tool.
Four.meme ignores us	Low	Medium	Mirage works with or without partnership; consumer product thesis stands alone.

Kill criteria (when to pivot or stop)

- **8 weeks post-hackathon**: <2,000 TG users despite organic effort → distribution thesis failing, consider paid-only B2B pivot.

- **12 weeks**: <20% of verdicts match labeled ground truth on held-out test set → reasoning quality insufficient; revisit agent design.
- **16 weeks**: Paid conversion <0.5% at 10K users → willingness-to-pay thesis failing; test different pricing or pivot to API-first.
- **6 months**: Adversary adaptation drops precision below 65% and feature additions don't recover → defensive posture unsustainable; likely need proprietary data feed.

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12. Success Metrics & Validation

North Star Metric

Trust Verdicts served per week — captures both usage (breadth) and the Mirage-native behavior (users consulting before copying), distinct from generic trading-bot metrics.

Input (leading) metrics

1. New Telegram bot users per week
2. Wallet subscriptions created per week
3. Screenshots of verdicts shared (trackable via UTM on share link)
4. Average verdicts per user per day
5. Evidence-citation pass rate (LLM output quality)

Output (lagging) metrics

1. Paid Pro conversions
2. Precision on labeled adversarial set (target >85%)
3. User-reported "save rate" (post-copy survey: "did Mirage save you?")
4. Retention at D7 / D30 / D90
5. MRR

30 / 60 / 90 day targets

Metric	30d	60d	90d
TG users	2,000	6,000	12,000
Verdicts / week	10K	40K	100K
Precision on test set	80%	85%	90%
Paid Pro subs	0	50	250
Integrations / partners	0	1	3

Pre-build validation experiments

Experiment 1 — Manual adversary catch. Before agents exist, hand-label 30 "popular alpha" wallets from BNB Telegram groups. Build Google Sheet with forensic checks. Publish 5 "caught" screenshots. **Success criterion:** ≥3 get ≥500 impressions each, ≥1 is shared by the original shiller.

Experiment 2 — Pricing willingness-to-pay. Stripe waitlist at $20/mo with "launching when we hit 500 subscribers." **Success criterion:** ≥100 pre-orders by end of Week 2.

Experiment 3 — Telegram onboarding funnel. Ship a coming-soon bot that accepts wallet addresses and responds "we'll send you your Mirage report when we launch — submit your email." **Success criterion:**

≥1,000 wallets submitted in 2 weeks; ≥300 emails collected.

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13. MVP Development Timeline

Reconciled with the Four.meme AI Sprint build window (April 8–21, 2026) — submission April 22.

Week 1 — April 8–14: Foundations

- Stand up BscScan + Dune ingestion.
- Build core feature extractors (bundle coefficient, co-buyer Jaccard, timing entropy, latency, graph distance, cross-token variance).
- Label training/evaluation set (500 wallets from public scam DBs + known-legit).
- Ship REST `/score_wallet/:address` with raw features + baseline statistical score.

Week 2 — April 15–21: Agents, Interfaces, Ship

- Write Coin, Wallet, Timing agent prompts with few-shot CoT examples drawn from Luo et al.
- Orchestrate in LangGraph; wire to feature pipeline; implement evidence-citation validator.
- Build Telegram bot (grammY) with `/analyze wallet`, `/analyze token`, `/subscribe`.
- Build Next.js dashboard with token card + ranked buyer table + reasoning expanders.
- End-to-end evaluation: 20 known-bad + 20 known-good; record precision/recall.
- Record demo video (3 live catches: bundle bot, wash trade, verified alpha).
- Finalize README, pitch deck, submission.
- **Submit April 22.**

Post-hackathon milestones

- **Week 3–4 (April 22–May 5):** Ship wallet subscription + alert pings, creator self-audit flow, backtest page.
- **Week 5–8 (May 6–June 2):** Pro tier billing, onboarding polish, first integration partner.
- **Week 9–16 (June–July):** Auto-copy with session keys; Solana adapter.

Team requirements

Role	Hours / week (hackathon)	Post-hackathon
Full-stack / backend (data + agents + API)	40–50h	1 FTE
Frontend (Next.js + bot + UX)	30–40h	0.7 FTE
Product/GTM/content	15–20h	0.5 FTE
Total	2 builders + 0.5 support	2.2 FTE post-hackathon

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14. CEO/CTO Decision Brief

Build confidence: HIGH

Research is peer-reviewed and recent. Distribution channel (Telegram) is mature. Competitive gap is concrete and defensible. MVP scope is honest and shippable in two weeks.

Key assumptions that must be true

1. **Multi-agent CoT generalizes from Solana research to BNB on-chain features.** (Validated structurally — same manipulation patterns, same feature classes.)

2. **Evidence-cited reasoning is more trusted than black-box scores.** (Validated qualitatively via user research in crypto safety tools like RugCheck.)
3. **BNB memecoin traders will adopt a new third-party tool.** (Validated by GMGN/BullX/MEVX installed-base growth on BNB.)
4. **A free Telegram surface generates viral "AVOID" screenshots.** (Hypothesis; will be de-risked in validation experiment #1.)

Biggest unknown

Will adversaries adapt faster than Mirage can evolve features? De-risked by choosing structural features with high adaptation cost; further de-risked by the data moat — the product gets smarter with every query.

If we had only 2 weeks

We would ship exactly the hackathon P0 scope above. The timeline is not aspirational — it is already what the build plan is. The validation experiments (#1 and #3) run in parallel during the same two weeks.

Comparable exits / acquisitions

- **Nansen** — $750M valuation as wallet-intelligence (institutional).
- **Arkham** — public token, ~$200M+ FDV at peak.
- **Dune Analytics** — $1B valuation (Series B 2022).
- **Chainalysis** — $8.6B valuation (tooling for adversary detection).

The revenue-scale benchmark is Nansen ($50M+ ARR) for a premium wallet-intel product. Mirage targets the long tail Nansen ignores: retail memecoin traders, where the WTP per user is lower but the count is 100× larger.

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15. Appendix: Research Sources

Academic

- Luo et al., *"Resisting Manipulative Bots in Meme Coin Copy Trading: A Multi-Agent Approach with Chain-of-Thought Reasoning,"* arXiv 2601.08641, ACM WWW 2026 — abs / HTML / PDF / NASA ADS / summary

Four.meme / BNB Chain ecosystem

- Four.meme official
- BNB Chain memecoin landscape blog
- DappBay Four.meme listing
- DWF Labs BNB launchpad comparison
- KuCoin Learn: Four.meme guide

Hackathon

- Four.meme AI Sprint on DoraHacks
- Phemex coverage
- Metaverse Post coverage
- Four.meme announcement on X

Competitive intelligence

- GMGN AI 2026 guide
- Banana Pro vs Axiom vs GMGN vs BullX ranking 2026
- Top 11 Telegram trading bots March 2026

- GMGN vs Axiom Trade comparison April 2026
- Top 5 Telegram bots for BNB Chain April 2026

Key data points referenced

- Four.meme 812K DAU + $1B PCS volume (BNB Chain blog, Q1 2026)
- GMGN $100M+ daily volume, ~50% SOL memecoin share (2025)
- Axiom 0.75–0.95% per-trade pricing (2026)
- BullX ~1% per-trade pricing (2026)
- 6,000+ memecoin projects in Luo et al. dataset (WWW 2026)

End of PRD.